Stat 184 Free Throw Analysis

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1 Introduction

This report presents a data analysis and exploration of the question: “What makes a good NBA free throw shooter?” We used player height and position and examined their relationship to free throw percentage among players with at least 150 free throw attempts during the 2025-2026 regular season.

2 Data Provenance

2.1 Who Collected It

The data was manually collected and compiled by our team. Player statistics were gathered by hand from Databallr (databallr.com) and entered into an Excel spreadsheet, which was then saved as a CSV file for use in this analysis.

2.2 What the Data Is

Each case (row) in the dataset represents one NBA player during the 2025-2026 regular season. The attributes include player name, team, position, physical measurements (height and wingspan in inches), and performance statistics such as free throw percentage (FT%) and free throw attempts (FTA).

2.3 When It Was Collected

The data was manually recorded on 4/28/2026. The
statistics reflect the 2025-2026 NBA regular season as displayed on Databallr at the time of collection.

2.4 Where It Came From

Statistics were transcribed by hand from databallr.com into an Excel spreadsheet. The original source of the underlying data is the NBA’s official game and player records.

2.5 Why It Was Collected

The data was collected to investigate how player height and position relate to free throw shooting percentage.

2.6 How It Was Collected

Team members used the features on Databallr to retrieve the necessary statistics. manually looked up each player on Databallr and recorded into an Excel file. This file was then exported as a CSV for use in R.

3 FAIR Principles

- **Findable:** The original statistics are publicly viewable on databallr.com. Our compiled dataset is stored in our team's GitHub repository.
- **Accessible:** The CSV file is available in our repo and can be downloaded by anyone with access to it.
- **Interoperable:** The CSV format is compatible with many coding languages
- **Reusable:** Column names are clearly labeled and the data context is well documented in this report, making it reusable for future analysis.

4 CARE Principles

This dataset covers NBA players whose statistics are part of the public record. No private or sensitive personal information was collected beyond what is officially published by the league.

5 Data Cleaning and Feature Creation

Before analysis, the dataset was cleaned and transformed. This included:

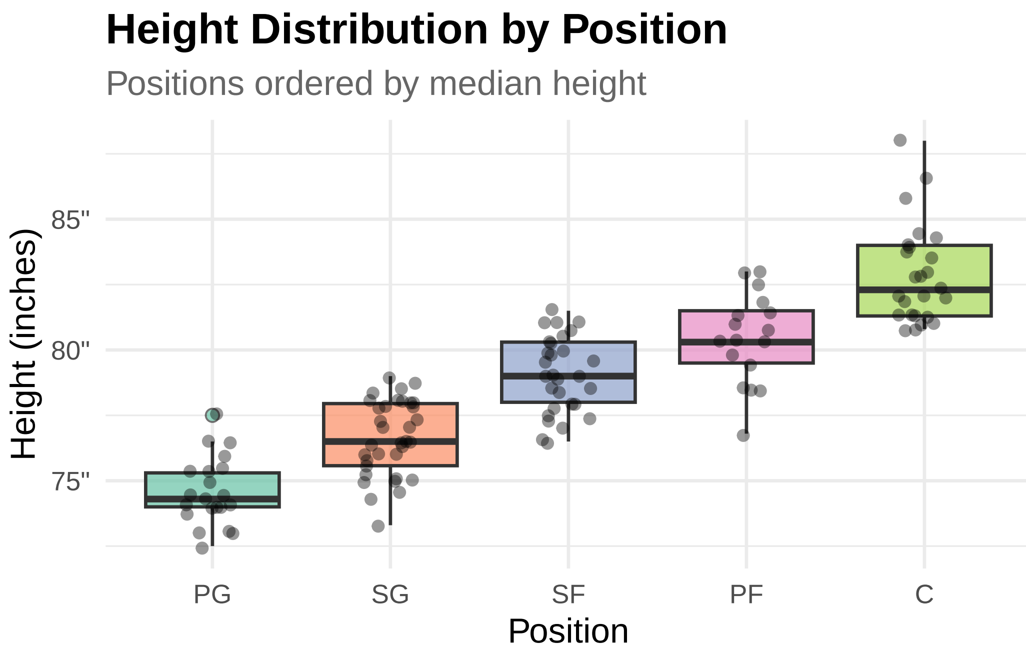
- Removing percentage symbols from free throw percentage
- Removing commas from total minutes
- Converting variables to numeric format

These transformations ensure that the data are consistent and suitable for analysis.

6 Exploratory Data Analysis

6.1 Robert

In this section, we look at if the height and position of the players and see if they correlate, which is the basis of our overall analysis.



Box plot showing NBA player height distribution by position, ordered by median height from shortest to tallest: PG (~74”), SG (~77”), SF (~79”), PF (~80”), and C (~82”). Centers show the greatest spread and tallest outliers; point guards are the shortest and most compact group.

Table 1: Height Summary by Position for Players Who Shot At Least 150 FT’s (tallest to shortest)

Position	Player Count	Avg Height (in)	Median Height (in)	Min Height (in)	Max Height (in)
C	25	82.9	82.3	80.8	88.0
PF	17	80.4	80.3	76.8	83.0
SF	29	79.0	79.0	76.5	81.5
SG	34	76.6	76.5	73.3	79.0
PG	21	74.6	74.3	72.5	77.5

The box plot and summary table together confirm a clear and consistent relationship between position and height in the 2025-2026 NBA season. From the table, we see that centers are the tallest position with an average height of 82.9 inches, while point guards are the shortest at an average of 74.6 inches, a difference of over 8 inches.

Looking at the box plot reveals that the positions follow a staircase pattern from PG to C, suggesting that height is a strong defining characteristic of positional assignment in the NBA. Also, most positions have relatively tight height distributions, meaning players at the same position tend to be similar in height. This clear correlation between height and position sets up our further research into how these attributes affect the free throw percentage of the players in our dataset.

6.2 Connor

This section examines whether a player's height is related to their free throw percentage within each NBA position. While Robert showed that height differs significantly across positions, this analysis focuses on whether taller or shorter players within the same role tend to shoot free throws more effectively. To evaluate this relationship, we calculate the correlation between height and free throw percentage for each position.

As shown in Figure 1, most positions exhibit correlations close to zero, indicating little to no linear relationship between height and free throw percentage within those groups. This suggests that among players who already share a similar role and body type, being slightly taller or shorter does not meaningfully impact free throw shooting ability.

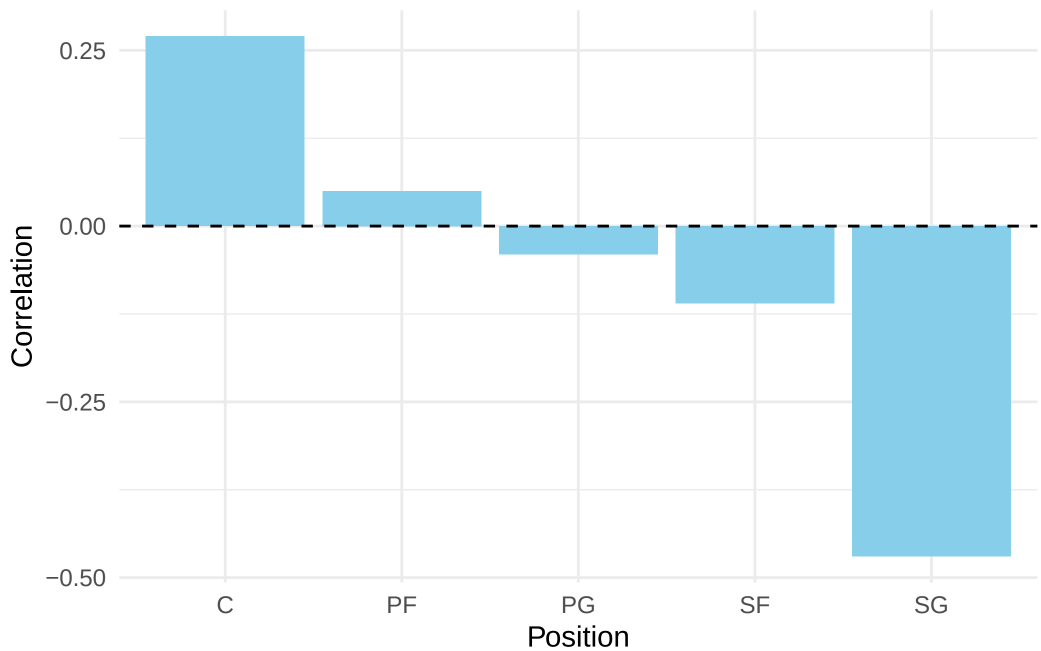


Figure 1: Correlation between height and free throw percentage by position.

Table 2 provides the corresponding numerical values. Centers show a modest positive correlation (0.27), while shooting guards show a more noticeable negative correlation (-0.47). However, because the results vary across positions and most correlations remain relatively weak, the overall pattern suggests that height alone is not a consistent predictor of free throw performance within a position group.

Table 2: Correlation between height and free throw percentage by position.

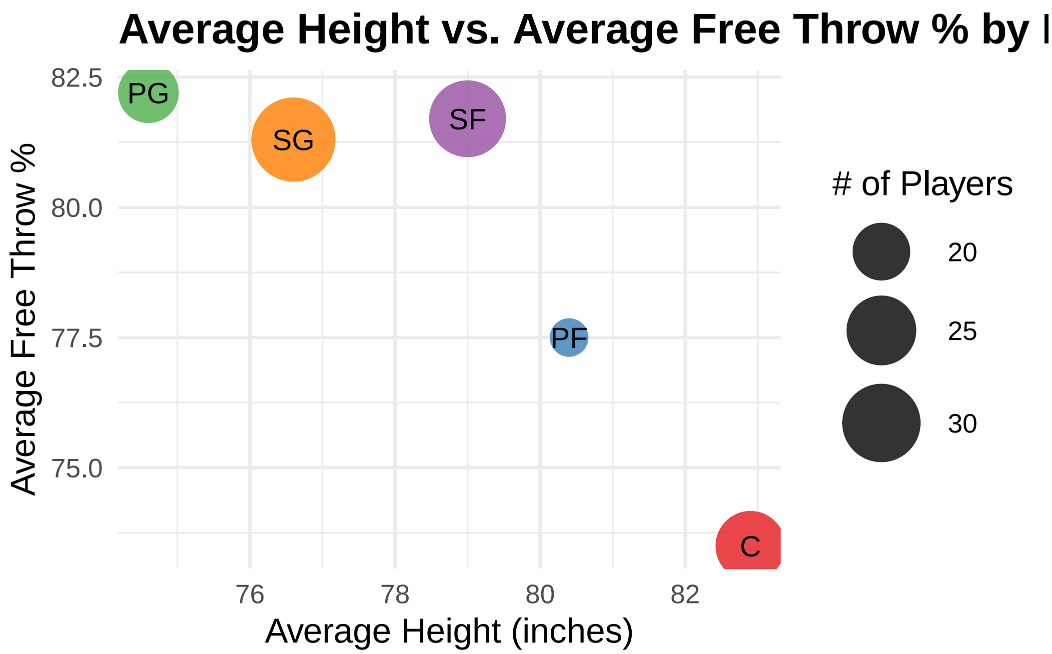
Position	Correlation	Avg FT%	Avg Height	Players
C	0.27	73.5	82.9	25
PF	0.05	77.5	80.4	17
PG	-0.04	82.2	74.6	21
SF	-0.11	81.7	79.0	29
SG	-0.47	81.3	76.6	34

These results complement Robert's findings by showing that although height is strongly associated with NBA position, it does not explain much of the variation in free throw percentage once players are compared within the same position. This also helps support Aidan's broader positional analysis, which examines how average free throw percentages differ across positions overall. Together, the analyses suggest that free throw shooting ability is driven more by individual skill and technique than by small differences in player height.

6.3 Aidan

Robert confirmed that height varies by position and Connor showed that height has little correlation with accuracy within each position, so I am zooming out and looking at the broader question: When we look at each position as a whole, which position tends to shoot free throws better on average? To explore this, I summarized the data by position and created a bubble chart where each bubble represents one position to show the correlation.

@plot-aidan makes the trend pretty easy to see. You can see how there is little difference in the average FT% in guards and small forwards, who also tend to shoot much more accurate than bigs, as they're all fairly even on the Y-axis. The scatter plot helps show a pretty consistent drop off in FT% as height rises, shown by about a 5% drop from small forwards to power forwards, and another 5% drop from power forwards to centers. We can see that players who are either point guards, shooting guards, or small forwards are on average 79 inches or shorter. From this we can make the observation that players in those positions typically have the ideal attributes to shoot a much more accurate FT% compared to the larger positions in power forwards and centers.



@table-aidan helps viewers clearly see how close the FT% really is between point guards, shooting guards, and small forwards, even given the signifigant average height difference from PG-SF. Having the actual numbers displayed in addition to the table really helps add emphasis on those three positions shooting within 1.1% of each other on average.

Table 3: Average Height and Free Throw Percentage by Position

Position	Avg Height (in)	Avg Free Throw %	# Players
PG	74.6	82.2	21
SF	79.0	81.7	29
SG	76.6	81.3	34
PF	80.4	77.5	17
C	82.9	73.5	25

7 Overall Conclusion

8 Open Science and Reproducibility

9 Author Contributions

- Robert Dang: The first portion of the QMD (marked Robert), GitHub repo setup, the introduction portion, data provenance, FAIR and CARE portions of the QMD
- Connor Box:
- Aidan Wieners:

10 AI Usage

Usage 1: Robert's alt text 05/06/26 GenAI tool used: Anthropic Claude | Claude Sonnet 4.6(claude.ai) Prompt: "create a short alt text for this data visualization involving nba player height and position" Exact Response: "Here's alt text for this visualization: Short alt text (for most contexts): Box plot showing NBA player height distribution by position, ordered by median height from shortest to tallest: PG (~74"), SG (~77"), SF (~79"), PF (~80"), and C (~82"). Centers show the greatest spread and tallest outliers; point guards are the shortest and most compact group."

Usage 2: Connor's alt text 05/06/26
GenAI tool used: OpenAI ChatGPT-5.5 (chatgpt.com)
Prompt: "create a short alt text for this data visualization involving nba player height and free throw percentage by position"
Exact Response: "Bar chart comparing the correlation between player height and free throw percentage across NBA positions (C, PF, PG, SF, SG). Most correlations are close to zero, indicating little relationship between height and free throw shooting within positions. Centers show a small positive correlation, while shooting guards show the strongest negative correlation."

11 Appendix: Code

Robert's code

```r
# Tidyverse Style Guide

# Step 1: Load packages
library(ggplot2)
library(dplyr)
library(readr)
library(ggrepel)
library(knitr)
library(kableExtra)

# Step 2: Load and tidy the data
df <- read_csv("NBAPlayerStats.csv") |>
  rename(
    ft_pct = `FT%`,
    height = `Height (in)`
  ) |>
  mutate(
    ft_pct = as.numeric(gsub("%", "", ft_pct))
  )

# Step 3: Summarize average height and FT% per position
position_summary <- df |>
  group_by(Position) |>
  summarise(
    avg_height = mean(height, na.rm = TRUE),
    avg_ft_pct = mean(ft_pct, na.rm = TRUE),
    count      = n()
  )

# Plot for height distribution by position
df |>
  mutate(Position = reorder(Position, height)) |>
  ggplot(aes(x = Position, y = height, fill = Position)) +
  geom_boxplot(alpha = 0.7, outlier.shape = 21, outlier.size = 2) +
  geom_jitter(width = 0.15, alpha = 0.4, size = 1.5) +
  scale_fill_brewer(palette = "Set2") +
  scale_y_continuous(labels = function(x) paste0(x, '"')) +
  labs(
    title    = "Height Distribution by Position",
    subtitle = "Positions ordered by median height",
    x        = "Position",
    y        = "Height (inches)"
```

```
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title      = element_text(face = "bold"),
    plot.subtitle   = element_text(color = "grey40"),
    legend.position = "none"
  )

# Table for height summary by position
df |>
  group_by(Position) |>
  summarise(
    `Player Count`        = n(),
    `Avg Height (in)`     = round(mean(height, na.rm = TRUE), 1),
    `Median Height (in)` = round(median(height, na.rm = TRUE), 1),
    `Min Height (in)`     = round(min(height, na.rm = TRUE), 1),
    `Max Height (in)`     = round(max(height, na.rm = TRUE), 1)
  ) |>
  arrange(desc(`Avg Height (in)`)) |>
  kable(
    caption = "Height Summary by Position for Players Who Shot At Least 150 FT's (tallest to
    align   = c("l", "c", "c", "c", "c", "c")
  ) |>
  kable_styling(
    bootstrap_options = c("striped", "hover", "condensed"),
    full_width        = FALSE
  )
```

Connor's code

```
### Author: Connor Box

# Step 1: Load packages
library(tidyverse)
library(knitr)

# Step 2: Load the NBA player data
nba_raw <- read_csv("NBAPlayerStats.csv")

# Step 3: Clean the data and rename columns
nba_clean <- nba_raw |>
  rename(
```

```
    position = Position,
    ft_percent = `FT%`,
    height_in = `Height (in)`
  ) |>
  mutate(
    ft_percent = as.numeric(str_remove(ft_percent, "%")),
    height_in = as.numeric(height_in)
  )

# Step 4: Calculate the correlation between height and free throw percentage by position
position_corr <- nba_clean |>
  group_by(position) |>
  summarise(
    correlation = round(cor(height_in, ft_percent, use = "complete.obs"), 2),
    avg_ft = round(mean(ft_percent, na.rm = TRUE), 1),
    avg_height = round(mean(height_in, na.rm = TRUE), 1),
    n_players = n(),
    .groups = "drop"
  )

# Step 5: Create a bar chart showing the correlations by position
#| label: fig-connor
#| fig-cap: "Correlation between height and free throw percentage by position."
#| fig-alt: "Bar chart comparing the correlation between player height and free throw percen
#| fig-pos: "H"

ggplot(position_corr, aes(x = position, y = correlation)) +
  geom_col(fill = "skyblue") +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(
    x = "Position",
    y = "Correlation"
  ) +
  theme_minimal()

# Step 6: Create a table showing the correlation values and summary statistics
#| label: tbl-connor
#| tbl-cap: "Correlation between height and free throw percentage by position."
#| tbl-pos: "H"

kable(
  position_corr,
```

```
  col.names = c("Position", "Correlation", "Avg FT%", "Avg Height", "Players")
)
```

Aidan's Code

```
### Author: Aidan Wieners
library(readr)
library(dplyr)
library(ggplot2)
library(knitr)

# Tidy the data
nba <- read_csv("NBAPlayerStats.csv") |>
  rename(
    ft_pct   = `FT%`,
    height   = `Height (in)`,
    position = Position
  ) |>
  mutate(
    ft_pct = as.numeric(gsub("%", "", ft_pct)),
    height = as.numeric(height)
  )

# Summarize by position
pos_summary <- nba |>
  group_by(position) |>
  summarise(
    avg_height = round(mean(height, na.rm = TRUE), 1),
    avg_ft_pct = round(mean(ft_pct,  na.rm = TRUE), 1),
    n_players  = n(),
    .groups = "drop"
  )

#Plot-aidan - Average height vs. FT% by position
#| label: plot-aidan
#| fig-cap: "Average height vs. average FT% by position. Bubble size shows number of players
#| fig-alt: "Scatter plot of average height vs. free throw percentage by basketball position

ggplot(pos_summary, aes(x = avg_height, y = avg_ft_pct,
                        size = n_players, color = position)) +
  geom_point(alpha = 0.8) +
  geom_text(aes(label = position), color = "black", size = 4, hjust = 0.5, vjust = 0.5)+
```

```r
  scale_size_continuous(range = c(6, 14), name = "# of Players") +
  scale_color_brewer(palette = "Set1", guide = "none") +
  labs(
    title = "Average Height vs. Average Free Throw % by Position",
    x = "Average Height (inches)",
    y = "Average Free Throw %"
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.title    = element_text(face = "bold")
  )

# Table-aidan - Average height vs. FT% by position

#| label: tab-aidan
#| fig-cap: "Provides the corresponding numerical values (Position, Avg Height, Avg FT%, # of
#| fig-alt: "Table showing average height, free throw percentage, and number of players for 
pos_summary |>
  arrange(desc(avg_ft_pct)) |>
  kable(
    caption = "Average Height and Free Throw Percentage by Position",
    col.names = c("Position", "Avg Height (in)", "Avg Free Throw %", "# Players"),
    align = c("l", "c", "c", "c")
  )
```